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KDCS-PPI: Knowledge distillation with counterfactual sampling for Protein-Protein Interaction prediction

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ABSTRACT

As the core of various biochemical reactions in life, Protein-Protein Interactions (PPIs) play a crucial role in maintaining the homeostasis of cellular functions, making the accurate prediction of PPIs particularly important. Traditional wet lab methods for predicting PPIs are time-consuming and costly. In contrast, PPI prediction methods utilizing Graph Neural Networks (GNNs) have exhibited promising performance and have increasingly emerged as the predominant approach in recent years. While GNNs rely on neighbor message aggregation, which can result in computational inefficiencies, Multilayer Perceptron (MLP) stand out for their time efficiency, as they do not require intricate handling of relational knowledge. However, MLPs often exhibit comparatively lower prediction accuracy. To leverage the advantages of both GNNs and MLPs in terms of effectiveness and efficiency, knowledge distillation techniques can be used to transfer the knowledge learned by GNNs to MLPs. During the knowledge distillation process, the knowledge transfer usually involves node feature embeddings rather than the interaction relationship knowledge between PPIs. Moreover, current methods frequently choose positive and negative samples for anchor nodes via random sampling, leading to suboptimal accuracy, especially for negative samples. To address this, we propose ***K**nowledge **D**istillation with **C**ounterfactual **S**ampling for **P**rotein-**P**rotein **I**nteraction prediction* (KDCS-PPI). Our method facilitates the transfer of diverse relational knowledge between proteins during the knowledge distillation process and utilizes a counterfactual sampling strategy to select more pertinent positive and negative examples. Extensive experiments on three datasets demonstrate that KDCS-PPI can be applied to large-scale PPI prediction tasks and achieves significant improvements in both effectiveness and computational efficiency compared to other benchmark methods. Our source codes will be publicly available at https://github.com/bin-db/KDCS-PPI.

1. Introduction

Protein-Protein Interactions (PPIs) (Hu, Wang, et al., 2021; Lee, 2023) are fundamental in the life sciences, representing the formation of protein complexes by two or more protein molecules through non-covalent bonds. These interactions form the backbone of the cellular biochemical reaction network and are essential for various life processes. Predicting PPIs holds significant importance for drug design and disease treatment. While experimental techniques (Fields & Song, 1989; Ho et al., 2002; Zhou et al., 2016) like Fluorescence Resonance Energy Transfer (FRET) (Clegg, 1995) and Bimolecular Fluorescence Complementation (BiFC) (Kodama & Hu, 2012) have been proposed for high-throughput PPI prediction over the past decades, their application is hindered by high costs and time requirements. As a result, there is a growing demand for efficient computational methods to predict unknown PPIs on a large scale.

In recent years, advancements in deep learning technologies have revolutionized PPI prediction tasks (Gao et al., 2023; Wu et al., 2024; Zhao et al., 2023). Typically, the process involves deriving initial protein representations from sequence or structural data (Hashemifar et al., 2018; Zhang, Xu, et al., 2023). These proteins are then represented as nodes within a PPI graph, where Graph Neural Networks (GNNs) are deployed to refine these protein representations via iterative message-passing mechanisms. The updated representations of the target protein nodes are fused and fed into a predictor to determine whether an interaction exists between them. The effectiveness of GNNs in PPI prediction can be attributed to their ability to assimilate contextual information from neighboring nodes (Xie et al., 2020). Nevertheless, this

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reliance on methods for obtaining and aggregating neighbors can lead to an exponential increase in the number of neighboring nodes during aggregation. Consequently, GNNs may incur substantial time overheads during both training and inference stages, posing scalability challenges for GNN-based PPI prediction methodologies, especially when dealing with large-scale PPI datasets.

Compared to graph structures, tabular data lacks complex topological structures, making it highly efficient to process (Guo et al., 2023). Therefore, Multi-Layer Perceptrons (MLPs) operating on tabular data consume significantly less time during training and inference compared to GNNs. To address the high computational cost and low time efficiency of GNNs, some recent works have started exploring how to combine GNNs with MLPs to leverage the effectiveness of GNNs while achieving the efficiency of MLPs. To this end, recent efforts have proposed using knowledge distillation techniques (Hinton, 2015; Hu, You, et al., 2021; Zhang, Liu, et al., 2021) to transfer knowledge learned by teacher GNNs to student MLPs. However, most of these works have focused primarily on node-level or graph-level tasks, with limited attention given to node-to-node link relationships. This makes it challenging to apply such approaches to PPI prediction tasks that are more concerned with link relationships.

In the knowledge distillation process, current methods commonly rely on random sampling to choose positive and negative samples for anchor nodes (Guo et al., 2023). Nonetheless, this method frequently generates imprecise samples, particularly for negative instances. This challenge stems from two main factors: Firstly, the validation and test sets contain interactions that are not exposed during training. This arrangement may result in positive samples for anchor nodes being mistakenly identified as negative samples, thereby degrading the model's accuracy. Secondly, undetected or undocumented interactions might be incorrectly labeled as negative samples in the modeling phase. This misclassification can lead to errors in the model's learning, ultimately limiting its overall performance.

The process of transferring knowledge acquired by GNNs from PPI network to MLPs via knowledge distillation poses two primary challenges: (1) *How to explicitly capture knowledge concerning protein-protein interaction relationships rather than focusing solely on node-level or graph-level representations?* (2) *How to select appropriate positive and negative samples for anchor nodes during the distillation process to enhance model performance?*

To tackle the above two challenges, we introduce a novel method named Knowledge Distillation with Counterfactual Sampling for Protein-Protein Interaction prediction (KDCS-PPI). To overcome the difficulty of transferring relational knowledge, we follow the approach of LLP (Guo et al., 2023) and employ two relational knowledge distillation strategies during the distillation process: a ranking-based matching strategy and a distribution-based matching strategy. Unlike previous works that attempt to align node representations in the teacher and student networks or match the logits scores of link probabilities between randomly selected node pairs, our proposed relational knowledge distillation strategies focus on the link probability relationships between each target protein (anchor node) and its positive/negative protein samples. Specifically, the ranking-based matching strategy emphasizes the ranking of the logits of link probabilities between positive/negative protein samples and the target protein, while the distribution-based matching strategy focuses on the distribution of the logits values of link probabilities between different samples and the target protein.

Meanwhile, to address the issue of unreliable samples, we employ counterfactual reasoning to enhance the sampling process. Counterfactual reasoning is a causal reasoning method used to determine what would have happened if conditions were different. Its core idea is to compare the observed outcomes in the real world with the potential outcomes that might arise if certain variables were altered. It involves three key elements: (1) **Context**, which refers to the background information influencing counterfactual reasoning; (2) **Treatment**, which represents the intervention or manipulation applied to an entity; and (3) **Outcome**, which denotes the observed or predicted outcome under a given context and treatment. In the proposed counterfactual sampling strategy, protein node representations serve as the context, modifications to the graph structure are considered as the treatment, and the existence of links between protein nodes is viewed as the outcome. Under the existing graph structure (Treatment = 1), the observed presence or absence of node links (Outcome = 0 or 1) represents the factual scenario. In contrast, after perturbing the graph structure (Treatment = 0), the observed link presence status represents the counterfactual scenario. The objective of the counterfactual sampling strategy is to select nodes that either maintain or lose links with the anchor node in both factual and counterfactual scenarios, designating them as positive or negative samples, respectively. We argue that, compared to randomly selecting positive and negative samples, this approach yields more reliable samples, thereby benefiting the model's training process.

In summary, the contributions of this paper are as follows:

- Our model prioritizes knowledge regarding protein-protein interaction relationships during the knowledge distillation process, enhancing the suitability of the student MLPs for PPI prediction task.
- Through the counterfactual sampling strategy, proteins that maintain the same link status with the target protein in both factual and counterfactual scenarios are selected as positive/negative samples, which is more beneficial for model training.
- Extensive experimental results validate that the proposed model outperforms or matches other baseline methods, striking a better balance between effectiveness and efficiency in PPI prediction.

The remainder of this paper is organized as follows: Section 2 reviews related work on KDCS-PPI. Section 3 introduces the symbols and concepts relevant to this study. Section 4 provides a detailed description of the components included in the proposed model. Section 5 presents the experiments designed to validate the model and discusses the results. Finally, Section 6 summarizes the characteristics of the model and outlines directions for future research.

2. Related work

This section will introduce the relevant content and recent advances in four research directions related to KDCS-PPI: protein-protein interactions, knowledge distillation for GNNs, link prediction, and counterfactual reasoning.

2.1. Protein-Protein Interactions (PPIs)

PPI determination (Balogh et al., 2022; Hu, Wang, et al., 2021; Lee, 2023) is crucial in the field of biomedicine, as understanding PPI is key to decoding the dynamics of complex biological systems and uncovering potential therapeutic intervention pathways. Traditional wet-lab approaches (Clegg, 1995; Kodama & Hu, 2012) for analyzing PPIs are costly and time-consuming, which has led to the development of various deep learning methods for PPI prediction. Current methods for PPI prediction generally follow two main steps: (1) Encoding protein information into embeddings, and (2) Inferring unknown PPIs using existing PPI data. The prevalent method integrates known PPI relationships into a PPI network and utilizes GNNs to glean insights from this network to predict unknown PPIs. For the first step, two major approaches are currently used: sequence-based (Kalifa et al., 2024; Lv et al., 2021; Zhao et al., 2023) and structure-based (Zhang, Xu, et al., 2023) methods. Since protein function is largely determined by protein structure, structure-based methods typically outperform sequence-based ones. However, in structure-based methods, HIGH-PPI (Gao et al., 2023) uses a dual-view hierarchical graph to learn protein representations. While this approach is effective, the hierarchical graph requires substantial

computational resources, resulting in low time efficiency for the model. Besides, methods like MAPE-PPI (Wu et al., 2024) accelerate the training process by pretraining to generate a "microenvironment codebook". However, since GNNs are still used for inference, the time efficiency remains suboptimal.

2.2. Knowledge Distillation for GNNs

Knowledge Distillation (Hinton, 2015) aims to transfer knowledge from a complex teacher model with numerous parameters to a simpler, lightweight student network with fewer parameters. Many studies have demonstrated that knowledge distillation can significantly reduce model complexity with minimal performance loss (Furlanello et al., 2018; Park et al., 2019). Due to GNNs' heavy reliance on neighborhood aggregation, which results in substantial computational overhead, and the fact that MLPs are highly efficient for non-structured data, there has been growing interest in transferring knowledge from GNNs to MLPs (Zheng et al., 2022). For example, Chen et al. (2022) propose SA-MLP, which introduces an innovative structure-mixing knowledge distillation approach to improve the capacity of MLPs in learning structural information. Zhang, Liu, et al. (2021) utilize the soft logits produced by the teacher GNNs to guide the training of the student MLP, resulting in impressive performance on node classification tasks. However, these distillation methods primarily focus on node-level or graph-level tasks, neglecting the relational knowledge critical for PPI prediction tasks, where the interaction between two proteins, rather than individual nodes or whole graphs, is of primary interest.

2.3. Link Prediction

Link Prediction (Le et al., 2023; Zhang & Chen, 2018) aims to predict the probability of a link between two nodes in a given network. This task has seen rapid developments in fields such as physics and computer science, and it has been widely applied across various domains (Li et al., 2017; Su et al., 2019; Yuen & Jansson, 2020). As a result, the link prediction problem has been the focus of extensive research. For example, Cai and Ji (2020) propose a novel node aggregation method for graph structures that converts closed subgraphs into different scales while preserving the relationships between two target nodes for link prediction. Liu et al. (2023) introduce a method combining GNNs with capsule networks for enhanced link prediction. Sun et al. (2024) leverage causal models by using node-pair information as context and global graph structure as input, with the existence of a link treated as the outcome. This approach enhances the model's ability to learn accurate representations for link prediction. Despite these advancements, no existing method has applied link prediction techniques specifically for PPI prediction. This gap highlights the potential for further exploration in combining link prediction methodologies with PPI networks to improve interaction prediction accuracy.

2.4. Counterfactual reasoning

Counterfactual reasoning is a causal reasoning method used to deduce what would happen if the conditions were different. Its core idea is to compare the observed outcomes in the real world with the potential outcomes that might occur if certain variables were to change. In the field of deep learning, counterfactual methods have been widely applied in areas such as recommendation systems (Li et al., 2024), anomaly detection (Xiao et al., 2024), and knowledge tracing (Zhang, Zhu, et al., 2023). For example, Zhao et al. (2022) used a causal model where the information of node pairs serves as context, global graph structural information as treatment, and the existence of links as the result, thereby enhancing graph data representation learning and improving the accuracy of link prediction. Yang, Chen, et al. (2023) employed counterfactual mechanisms to generate artificially difficult negative samples for graph contrastive learning, thus enhancing the effectiveness of graph learning. Additionally, Yang, Ma, et al. (2023) set the most recent practice answer as the counterfactual of the actual answer, using this counterfactual answer to evaluate the student's knowledge mastery. However, to date, there has been little research applying counterfactual methods to positive and negative sample sampling in PPI prediction.

3. Preliminaries

This section will introduce the problem definition, as well as the symbols required for knowledge distillation and counterfactual sampling, with the symbols organized in Table 1.

Given a set of proteins and their interactions, a PPI graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ can be constructed, where each node $v_i \in \mathcal{V}$ represents a protein, and $\mathcal{E} \in \mathcal{V} \times \mathcal{V}$ denotes the set of observed links (interactions). Let $\mathbf{A} \in \{0,1\}^{N\times N}$ represent the adjacency matrix of the PPI graph $\mathcal{G}$, where $\mathbf{A}_{i,j} = 1$ if there is a link between nodes v_i and v_j, and $\mathbf{A}_{i,j} = 0$ otherwise. The matrix $\mathbf{X} \in \mathbb{R}^{N\times F}$ represents the initial feature matrix of the N protein nodes, obtained through pretraining using the encoding book. Each row $\mathbf{x}_i$ corresponds to the F-dimensional feature vector of node v_i.

Following the widely accepted problem definition for link prediction in graph data (Cai et al., 2021; Zhang & Chen, 2018; Zhang, Li, et al., 2021), we define the PPI prediction task as predicting whether a link exists between each pair of target protein nodes, based on an observed PPI graph where validation and test links are masked.

Knowledge distillation (Hinton, 2015) aims to transfer knowledge from a large, complex teacher model to a smaller, simpler student network. During the knowledge transfer process, node features and prediction results learned by both the teacher and student models are involved. Let the node feature matrices learned by the teacher model and student model be denoted as $\mathbf{H} \in \mathbb{R}^{N\times D}$ and $\hat{\mathbf{H}} \in \mathbb{R}^{N\times D}$, where D represent the feature dimensions. Here, $\mathbf{h}_i$ and $\hat{\mathbf{h}}_i$ represent the features of node v_i, updated by the teacher model and student model, respectively. Similarly, let $y_{i,j}$ and $\hat{y}_{i,j}$ denote the predicted probability of a link existing between node pair (v_i, v_j) by the teacher GNNs and student MLP, respectively.

In the process of relational knowledge distillation (see Section 4.1), positive and negative samples need to be selected for each node. According to the counterfactual sampling strategy (see Section 4.2), we first construct a treatment matrix $\mathbf{T} \in \{0,1\}^{N\times N}$, indicating whether nodes are in the same community, and a similarity matrix $\mathbf{S} \in \mathbb{R}^{N\times N}$, representing the similarity between nodes. We also define a counterfactual treatment matrix $\mathbf{T}^{CF} \in \{0,1\}^{N\times N}$ and an adjacency matrix $\mathbf{A}^{CF} \in \{0,1\}^{N\times N}$ for the counterfactual scenario. Based on these, we compute the candidate positive and negative sample index matrices and the sampling probability matrices, denoted as $\mathbf{M} \in \mathbb{N}^{N\times 2k}$ and $\mathbf{P} \in \mathbb{R}^{N\times 2k}$, respectively, where k represents the number of candidate positive/negative samples. Furthermore, the index matrix and probability matrix for the candidate positive samples are denoted as $\mathbf{M}^{pos} \in \mathbb{N}^{N\times k}$ and $\mathbf{P}^{pos} \in \mathbb{R}^{N\times k}$, respectively. Similarly, the index matrix and probability matrix for negative samples are denoted as $\mathbf{M}^{neg} \in \mathbb{N}^{N\times k}$ and $\mathbf{P}^{neg} \in \mathbb{R}^{N\times k}$, respectively. During the model optimization process, based on the sampling probability matrices $\mathbf{P}_i^{pos}$ and $\mathbf{P}_i^{neg}$ for the anchor node v_i, we sample n positive samples from the candidate sample index matrix $\mathbf{M}_i^{pos}$ and n negative samples from $\mathbf{M}_i^{neg}$, respectively. These sampled instances form the sample set $C_i \in \mathbb{N}^{2n}$, which contains both positive and negative samples associated with the anchor node v_i. Finally, we use $\mathcal{Y}_i = \{y_{i,j} | v_j \in C_i\}$ and $\hat{\mathcal{Y}}_i = \{\hat{y}_{i,j} | v_j \in C_i\}$ to represent the sets of link probabilities between the samples in C_i and the anchor node v_i, as predicted by the teacher GNNs and the student MLP.

Table 1
Summary of symbols involved in the model.

Symbol	Description
$\mathcal{G}$	PPI graph
$\mathcal{V}$	Node set
$\mathcal{E}$	Edge set
$\mathbf{H}, \hat{\mathbf{H}}$	Feature matrices updated by the teacher GNNs and the student MLP, respectively
$\mathbf{h}, \hat{\mathbf{h}}$	Feature representations of nodes updated by the teacher GNNs and the student MLP
$\mathcal{Y}_i$, $\hat{\mathcal{Y}}_i$	Sets of predicted link probabilities between the samples in C_i and the anchor node v_i by the teacher GNNs and student MLP
$y_{i,j}, \hat{y}_{i,j}$	Predicted probabilities of a link between node pair v_i and v_j by the teacher GNNs and student MLP
$\mathbf{A}, \mathbf{A}^{CF}$	Adjacency matrix in the real scenario and in the counterfactual scenario
$\mathbf{T}, \mathbf{T}^{CF}$	Treatment matrix in the real scenario and in the counterfactual scenario
$\mathbf{P}_i, \mathbf{P}_i^{pos}, \mathbf{P}_i^{neg}$	Sampling probability matrices for candidate, positive, and negative samples for the anchor node v_i
$\mathbf{M}_i, \mathbf{M}_i^{pos}, \mathbf{M}_i^{neg}$	Index matrices for candidate, positive, and negative samples for the anchor node v_i
C_i	Sample set for node v_i

4. Methodology

In this section, we first provide an overview of the model, and then detail the key components involved in each section. Specifically, Section 4.1 introduces the two relational distillation strategies used in the model. Section 4.2 describes the counterfactual sampling strategy employed during model training. Finally, Section 4.3 presents the method for predicting PPIs and the overall training loss.

KDCS-PPI primarily involves four main components: (1) As shown in Fig. 1(a), following the approach outlined in MAPE-PPI (Wu et al., 2024), use the sequence and structural information (generated by AlphaFold2) of each protein in the PPI network to construct a corresponding graph for each protein, where the nodes represent amino acids and each edge represents the sequence and spatial location information between amino acids. Using the constructed protein graph information, a pre-trained codebook is generated. The obtained codebook is used to encode each protein into $\mathbf{x}_i$, resulting in the representation matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$ of proteins, which serves as the initial node features in the PPI graph. (2) As depicted in Fig. 1(b), the teacher GNN network is first trained using the structure and node features of the PPI graph. Then, a knowledge distillation framework is employed as the model's backbone. By utilizing two relational distillation strategies (ranking-based matching and distribution-based matching), the relational knowledge learned by the teacher GNN from the PPI graph's topological information and node features is transferred to the student MLP network, improving the inference performance of the student MLP. Taking anchor node v_2 as an example (as shown in the diagram), both the teacher GNN and the student MLP predict the link probabilities between each sample in the set of samples $C_2 = \{v_8, v_9, v_6, v_3\}$ and anchor node v_2. The corresponding logits are denoted as $\mathcal{Y} = \{y_{2,8}, y_{2,9}, y_{2,6}, y_{2,3}\}$ and $\hat{\mathcal{Y}} = \{\hat{y}_{2,8}, \hat{y}_{2,9}, \hat{y}_{2,6}, \hat{y}_{2,3}\}$, respectively. The goal of the ranking-based matching strategy is to align the ranking of all predicted values in the student MLP's prediction $\hat{\mathcal{Y}}$ with the ranking of the predictions in the teacher GNN's result $\mathcal{Y}$. The distribution-based matching strategy, on the other hand, focuses on aligning the distribution of the logits in the student MLP's prediction $\hat{\mathcal{Y}}$ with the distribution of the teacher GNN's prediction $\mathcal{Y}$. (3) In the knowledge distillation process, it is necessary to select reliable positive/negative samples for the anchor node. As shown in Fig. 1(c), first, a specific graph mining/clustering method (there are various graph mining/clustering methods available, and in this paper, we have employed the k-core algorithm (Seidman, 1983)) is applied to partition the nodes in the graph into different communities. This is represented by $\mathbf{T}_{i,j} = 1$, meaning v_i and v_j belong to the same community, and $\mathbf{T}_{i,j} = 0$ otherwise. As illustrated in the figure, we take the anchor node v_2 and the negative sample node v_8 as an example to explain the counterfactual sampling process. In the factual scenario ($\mathbf{T}_{2,8} = 0$), the link between v_2 and v_8 can be directly observed, i.e., $\mathbf{A}_{2,8} = 0$. However, in the counterfactual scenario ($\mathbf{T}_{2,8} = 1$), the existence of a link between v_2 and v_8 cannot be directly observed. Therefore, we use nodes v_9 and v_7, which are the most similar in representation (context) to nodes v_2 and v_8, but belong to opposite communities (i.e., opposite treatment, $\mathbf{T}_{9,7} \neq \mathbf{T}_{2,8}$), to replace the existence of the link between v_2 and v_8 in the counterfactual scenario, i.e., $\mathbf{A}^{CF}_{2,8} = \mathbf{A}_{9,7} = 0$. Therefore, in both the factual and counterfactual scenarios, no link exists between v_8 and anchor node v_2, making v_8 a reliable negative sample for anchor node v_2. The process of selecting positive samples is analogous to that of negative samples. However, unlike negative samples which are chosen based on the absence of a link, positive samples are selected by identifying nodes that maintain a connection to the anchor node in both the factual and counterfactual scenarios. (4) The total loss used for model optimization.

The remainder of this section will provide a detailed explanation of (2) the relational knowledge distillation strategy, (3) the counterfactual node selection strategy, and (4) the total loss for the model.

4.1. Relational knowledge distillation strategies

This section will provide a detailed description of the relational knowledge distillation strategies used in the model, specifically the rank-based matching strategy and the distribution-based matching strategy. The goal of both matching strategies is to align the predictions of the student MLP and the teacher GNN regarding the link probabilities between a specific anchor node and its samples. However, the former focuses more on the ranking relationship between these predicted values, while the latter emphasizes the distributional relationship between the predicted values.

4.1.1. Rank-based matching strategy

Differences in the strength of interactions between proteins exist due to varying affinities between different protein pairs (Balogh et al., 2022; Nooren & Thornton, 2003). To enable the model to capture the strength of interactions between protein pairs, we first use the ranking-based matching strategy. Specifically, for anchor node v_i, the teacher GNN derives the logit value $y_{i,j}$ for each positive/negative sample v_j in the sample set C_i, resulting in $\mathcal{Y} = \{y_{i,j} | v_j \in C_i\}$. The corresponding predicted values from the student MLP are $\hat{\mathcal{Y}}_i = \{\hat{y}_{i,j} | v_j \in C_i\}$. Then, Eq. (1) is used to align the ranking of the values in $\hat{\mathcal{Y}}_i$ with those in $\mathcal{Y}_i$:

$$\mathcal{L}_{rank} = \sum_{v_i \in \mathcal{V}} \sum_{\{\hat{y}_{i,a}, \hat{y}_{i,b}\} \in \mathcal{Y}_i} \max(0, -r(\hat{y}_{i,a} - \hat{y}_{i,b}) + \delta),$$
$$\text{where} \quad r = \begin{cases} 1, & \text{if } y_{i,a} - y_{i,b} > \delta, \\ -1, & \text{if } y_{i,a} - y_{i,b} < -\delta, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

where, δ is a margin hyperparameter, typically a very small value. Introducing $r = 0$ in the formula, i.e., $|y_{i,a} - y_{i,b}| < \delta$, enables the student MLP to effectively distinguish between the minor logit value differences caused by noise in the teacher GNNs.

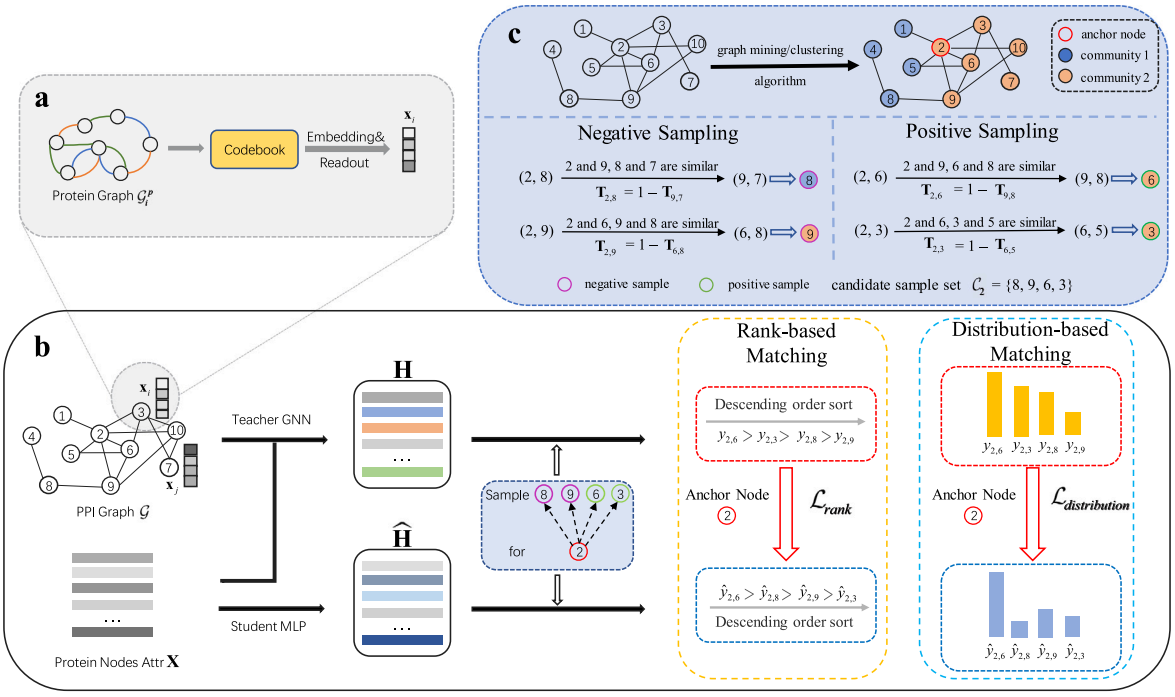


Fig. 1. Overview of KDCS-PPI. (a) Protein encoding method, using a pretrained CodeBook to encode proteins. (b) Knowledge distillation framework, transferring knowledge learned by the teacher GNN to the student MLP, employing two relational distillation strategies: rank-based matching and distribution-based matching. (c) Counterfactual sampling strategy, selecting positive/negative samples for the anchor node.

4.1.2. Distribution-based matching

As a coarse-grained ranking approach, the aforementioned ranking-based matching strategy effectively transfers the ranking information of the link probabilities between each node in the sample set $\mathcal{C}_i$ and the anchor node v_i. However, it fails to fully leverage the score information in the teacher GNN's predictions $\mathcal{Y}_i$. Therefore, the model also employs a distribution-based matching strategy. Unlike previous works (Furlanello et al., 2018; Hinton, 2015; Phuong & Lampert, 2019; Yan et al., 2020) that focus solely on a logits-matching strategy for independent node pairs, this strategy underscores the logit value distribution of link probabilities between each anchor node v_i and all nodes within its sample set $\mathcal{C}_i$. This approach highlights the link relationships between the anchor node and its positive/negative sample nodes.

Specifically, using Eqs. (2) and (3), we apply softmax normalization to the teacher GNN's predictions $\mathcal{Y}$ and the student MLP's predictions $\hat{\mathcal{Y}}$, converting them into probability distributions. The logit distribution of the teacher GNN is computed as follows:

$$P(\mathcal{C}_i) = \{\frac{exp(y_{i,a}/\tau)}{\sum_{b\in\mathcal{C}_i} exp(y_{i,b}/\tau)} | a \in \mathcal{C}_i\}, \tag{2}$$

where $P(\mathcal{C}_i)$ represents the logit value distribution of the teacher GNNs, $\mathcal{C}_i$ is the sample set for node v_i, and τ is a temperature hyperparameter used to control the softness of the softmax function. Similarly, the logit value distribution of the student MLP is computed as follows:

$$\hat{P}(\mathcal{C}_i) = \{\frac{exp(\hat{y}_{i,a}/\tau)}{\sum_{b\in\mathcal{C}_i} exp(\hat{y}_{i,b}/\tau)} | a \in \mathcal{C}_i\}. \tag{3}$$

Finally, we formulate the loss function $\mathcal{L}_{distribution}$ using KL divergence to align the student MLP's prediction distribution $\hat{P}(\mathcal{C}_i)$ with the teacher GNN's prediction distribution $P(\mathcal{C}_i)$, enabling a more fine-grained relational knowledge transfer, the specific formula is as follows:

$$\mathcal{L}_{distribution} = D_{KL}(P(\mathcal{C}_i) \| \hat{P}(\mathcal{C}_i)). \tag{4}$$

4.2. Counterfactual sampling strategy

In the process of model training, it is crucial to have a collection of nodes connected to the anchor node v_i, which we term positive samples, as well as a set of negative samples that are not connected to the anchor node. The quality of the negative samples plays a pivotal role in the effectiveness of the training process. Nonetheless, randomly selected negative samples can often be unreliable. This section begins by investigating the limitations of random negative sample selection through experimental analysis. Following this, we introduce the counterfactual sampling strategy that is utilized in our model to address these limitations.

4.2.1. Randomly select samples

As mentioned earlier, negative samples are crucial for model training, but randomly selected negative samples are often unreliable for two main reasons: (1) Some interactions are allocated to the validation and test sets and are thus not visible during training. As a result, the model might incorrectly sample these actual interaction relationships as negative samples during training. (2) The PPI network encompasses known interactions, while there may be unexplored interactions in the actual biological system. The purpose of prediction is to uncover these interactions. However, random negative samples might include these undiscovered interactions. Training the model with such interactions as negative samples could limit the model's predictive performance. Samples that arise from the above two scenarios are referred to as inaccurate samples.

Since the second scenario cannot be directly observed, we have only validated the first scenario. In the SHS27k dataset (see Section 5.1), various numbers of negative samples n are set for all nodes (1663 in total) in the PPI graph, and the number of inaccurate samples has been recorded in Fig. 2. It is observed that as the number of negative samples increases, the number of inaccurate samples also rises correspondingly. Note that Fig. 2 only displays the number of inaccurate samples from the first scenario. Considering both scenarios (1) and (2), the number of inaccurate samples might be even higher.

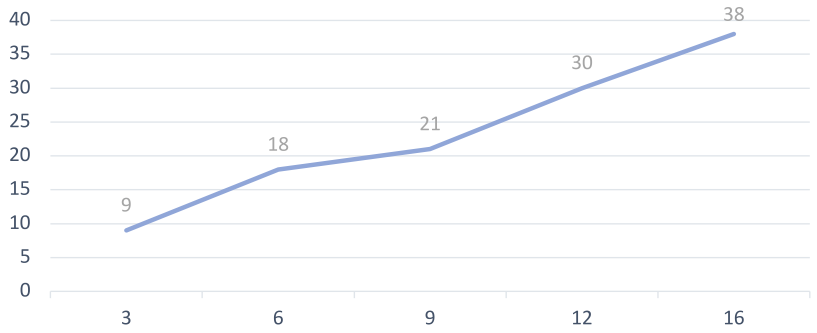
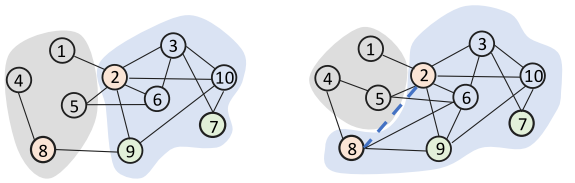


Fig. 2. Number of inaccurate nodes. For the 1663 nodes in the SHS27k dataset, n negative samples were randomly sampled. The x-axis represents the number of negative samples sampled for each node, while the y-axis represents the number of selected negative samples that appear in the validation and test sets, that is, the number of inaccurate samples.

The observations suggest that employing random negative sampling can lead to the model training on inaccurate negative samples. Furthermore, the model necessitates more dependable positive samples for effective training. We argue that, considering the possibility of data statistical errors in the real world, using nodes that have links with the anchor node in both factual and counterfactual scenarios as positive samples would be more reliable. Hence, a more rational approach is essential to select appropriate positive and negative samples for anchor nodes during training, enabling the student MLP to more effectively acquire relational knowledge from the teacher GNNs. As a result, we then introduce a counterfactual sampling strategy.



(a) Factual Scenario (b) Counterfactual Scenario

Fig. 3. Factual scenario and counterfactual scenario with structural changes. Different shades represent different community affiliations.

4.2.2. Counterfactual sampling strategy

In the real world, there are many proteins whose relationships with other proteins have not been fully explored. This is reflected in the PPI graph by the degree of these nodes being lower than that of other protein nodes. When selecting positive and negative samples, these factors should be excluded. Therefore, we aim to determine whether a specific protein still interacts with the anchor protein when the graph structure deviates from reality, as defined in the counterfactual scenario of this study. We argue that only nodes that consistently maintain or lack a link with the anchor node in both factual and counterfactual scenarios can be considered reliable positive or negative samples.

Inspired by recent research on counterfactual techniques applied to link prediction (Zhao et al., 2022), we propose a counterfactual sampling strategy to minimize the influence of irrelevant factors and select more accurate positive and negative samples during the sampling process.

As shown in Fig. 3(a), in the factual scenario, nodes v_2 and v_8 do not belong to the same community, and there is no link between them. However, if the interactions of v_8 are fully explored, as illustrated in Fig. 3(b), a graph mining/clustering method may partition v_2 and v_8 into the same community, which represents the counterfactual scenario. Our key question is whether, under this counterfactual scenario, v_2 and v_8 remain unlinked. If, in the counterfactual scenario, v_2 and v_8 remain unlinked, we consider v_8 to be a reliable negative sample for v_2. Conversely, if a link exists between them in the counterfactual scenario, v_8 is not considered a reliable negative sample for v_2. However, as we all know, the counterfactual scenario cannot be directly observed. Therefore, we approximate the counterfactual relationship between v_2 and v_8 by leveraging other node pairs that are most similar to them but belong to different communities. Suppose that nodes v_9 and v_7 are most similar to v_2 and v_8, respectively, and that they belong to the same community. In this case, the absence of a link between v_9 and v_7 can be used as an approximation of the counterfactual scenario where v_2 and v_8 are not linked. Consequently, v_8 can be considered a reliable negative sample for v_2. The sampling process for other negative and positive samples follows a similar approach. Following this process, we can select k positive and negative samples for anchor node v_2. To introduce randomness, we randomly select n positive and negative samples from the set of k candidates, with the selection probability based on the similarity between node pairs. The higher the similarity, the greater the probability of selection. For example, the sum of the similarities between the node pairs (v_2, v_9) and (v_8, v_7) is used as the basis for selecting v_8 as the final negative sample for anchor node v_2.

By employing this counterfactual sampling strategy, the impact of unreliable graph structural elements is mitigated, rendering the selected positive and negative nodes more reliable. For clarity, the samples selected using the counterfactual sampling strategy will be denoted as counterfactual samples. Subsequent sections will offer an elaborate account of the positive and negative sample selection process.

Calculate the treatment matrix T and similarity matrix S. Using a specific graph mining/clustering algorithm c, we partition the nodes in the PPI graph into different communities. Based on the community membership of each node, the processing matrix $\mathbf{T}$ is defined as follows:

$$\mathbf{T}_{i,j} = \begin{cases} 1, & \text{if } c(v_i) = c(v_j), \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

where $c(v_i)$ and $c(v_j)$ denote the community indices of nodes v_i and v_j respectively. $\mathbf{T}_{i,j}$ indicates whether nodes v_i and v_j belong to the same community. Meanwhile, we compute the similarity matrix between nodes, where $\mathbf{S}_{i,j}$ represents the similarity between nodes v_i and v_j. In this work, we adopt cosine similarity, the calculation is as follows:

$$\mathbf{S}_{i,j} = \frac{\mathbf{h}_i \cdot \mathbf{h}_j}{\|\mathbf{h}_i\| \|\mathbf{h}_j\|}, \tag{6}$$

where $\mathbf{S}_{i,j}$ represents the similarity between nodes v_i and v_j. Various similarity measures can be used, and in this work we adopt cosine similarity.

Construct the counterfactual treatment matrix $\mathbf{T}^{CF}$ and adjacency matrix $\mathbf{A}^{CF}$. For a node pair (v_i, v_j), we can only observe whether a link exists between them in the factual scenario. However, their link status in the counterfactual scenario (i.e., $\mathbf{T}^{CF}_{i,j} = 1 - \mathbf{T}_{i,j}$) remains unknown. Therefore, we need to identify an alternative node pair (v_a, v_b) that can serve as a proxy for (v_i, v_j) in the counterfactual setting. This substitute node pair must satisfy $T_{a,b} = 1 - T_{i,j}$ and have feature similarities with

v_i and v_j that exceed a predefined threshold. The detailed calculation process is as follows:

$$(v_a, v_b) = \underset{v_a, v_b \in \mathcal{V}}{arg\,max}\{\mathbf{S}_{i,a} + \mathbf{S}_{j,b} | \mathbf{T}_{a,b} = 1 - \mathbf{T}_{i,j}, \mathbf{S}_{i,a} + \mathbf{S}_{j,b} > 2\gamma\}, \tag{7}$$

where γ is a hyperparameter representing the minimum similarity between two pairs of nodes.

If a substitute node pair (v_a, v_b) can be identified via Eq. (7), it enables an approximate estimation of the link status for (v_i, v_j) in the counterfactual scenario. We define the counterfactual treatment matrix $\mathbf{T}^{CF}_{i,j}$ as the inverse of its factual counterpart, indicating that the link status of (v_i, v_j) can now be inferred in the counterfactual scenario. Meanwhile, we approximate the counterfactual adjacency matrix by assigning the link status of (v_a, v_b) to (v_i, v_j), i.e., $\mathbf{A}^{CF}_{i,j} = \mathbf{A}_{a,b}$. If no suitable substitute pair (v_a, v_b) is found, the counterfactual treatment and adjacency matrices remain the same as in the factual scenario. The specific calculation formula is as follows:

$$\mathbf{T}^{CF}_{i,j}, \mathbf{A}^{CF}_{i,j} = \begin{cases} 1 - \mathbf{T}_{i,j}, \mathbf{A}_{a,b}, & \text{if } \exists (v_a, v_b) \in \mathcal{V} \times \mathcal{V} \; satisfies \text{ Eq. (7)}, \\ \mathbf{T}_{i,j}, \mathbf{A}_{i,j}, & \text{otherwise.} \end{cases} \tag{8}$$

Constructing the candidate matrix $\mathbf{M}_i$ and the probability matrix $\mathbf{P}_i$ for positive/negative samples. For node pairs (v_i, v_j) that can approximate the counterfactual scenario using the above process, we can further compare their link consistency between the factual and counterfactual cases to determine whether v_j qualifies as a candidate positive or negative sample for v_i. Based on this, we construct the positive and negative sample sets for v_i. Considering the candidate positive sample set $\mathbf{M}^{pos}_i$, the calculation process is as follows:

$$\mathbf{M}^{pos}_i = \{v_j | \mathbf{T}_{i,j} = 1 - \mathbf{T}^{CF}_{i,j}, \mathbf{A}_{i,j} = \mathbf{A}^{CF}_{i,j} = 1\}, \tag{9}$$

where, $\mathbf{T}^{CF}_{i,j} = 1 - \mathbf{T}_{i,j}$ indicates that the counterfactual scenario for the node pair (v_i, v_j) can be approximated. Additionally, $\mathbf{A}^{CF}_{i,j} = \mathbf{A}_{i,j} = 1$ signifies that the link between v_i and v_j exists in both the factual and counterfactual scenarios. Through this approach, k positive samples are selected and added to $\mathbf{M}^{pos}_i$ as the candidate positive sample set for anchor node v_i. To introduce randomness, n samples are further selected from these k candidates as the final positive samples. The sampling probability is determined based on the similarity between node pairs, as defined in Eq. (7)—the higher the similarity, the greater the probability of selection. Taking the selection of v_j as the final sample for anchor node v_i as an example, the sampling probability $p_{i,j}$ is computed as follows:

$$p_{i,j} = \mathbf{S}_{i,a} + \mathbf{S}_{j,b}, \tag{10}$$

where, $\mathbf{S}_{i,a}$ represents the cosine similarity between nodes v_i and v_a. The sampling probability $p_{i,j}$ is processed through a softmax function and stored in $\mathbf{P}_i$, serving as the final sampling probability for each candidate positive sample.

$$\mathbf{P}^{pos}_i = softmax(\{p_{i,j} | v_j \in \mathbf{M}^{pos}_i\}). \tag{11}$$

If the number of positive samples that meet the counterfactual sampling constraints is fewer than k, other neighboring nodes of the anchor node v_i are used to supplement the candidate positive sample set. The corresponding sampling probabilities for these supplementary samples are made equal to each other but set to be lower than the sampling probabilities of the counterfactual samples.

The sampling process for candidate negative samples for the anchor node v_i is similar to that for positive samples:

$$\begin{aligned} \mathbf{M}^{neg}_i &= \{v_j | \mathbf{T}_{i,j} = 1 - \mathbf{T}^{CF}_{i,j}, \mathbf{A}_{i,j} = \mathbf{A}^{CF}_{i,j} = 0\}, \\ \mathbf{P}^{neg}_i &= softmax(\{p_{i,j} | v_j \in \mathbf{M}^{neg}_i\}). \end{aligned} \tag{12}$$

Select the final positive/negative sample sets. After obtaining $\mathbf{M}^{pos}$, $\mathbf{P}^{pos}$, $\mathbf{M}^{neg}$ and $\mathbf{P}^{neg}$, these four matrices are used to determine the final positive and negative samples for the anchor node v_i during training. First, randomly sample n (n<k) values according to the probability values in $\mathbf{P}^{pos}_i$ and $\mathbf{P}^{neg}_i$, respectively. Then, find the nodes corresponding to the indices in $\mathbf{M}^{pos}_i$ and $\mathbf{M}^{neg}_i$ to establish the positive and negative sample sets C_i for the anchor node v_i.

4.3. Overall optimization

During training, the MLP receives the features $\hat{\mathbf{h}}_i$ and $\hat{\mathbf{h}}_j$ for the node pair (v_i, v_j), these features are concatenated and considered as input to the predictor. The output of the predictor suggests the link prediction value:

$$\hat{y}_{i,j} = FC(concat(\hat{\mathbf{h}}_i | \hat{\mathbf{h}}_j)), \tag{13}$$

where $concat(\cdot|\cdot)$ denotes the concatenation operation. $\hat{y}_{i,j}$ represents the model's final prediction of the probability of the link between nodes v_i and v_j. The binary cross-entropy loss is then used to compute the loss against the true labels $\mathcal{L}_{sup}$:

$$\mathcal{L}_{sup} = -\frac{1}{N} \sum_{v_i \in \mathcal{V}} \sum_{v_j \in C_i} (y^t_{i,j} \log_2 (\hat{y}_{i,j}) + (1 - y^t_{i,j}) \log_2 (1 - \hat{y}_{i,j})), \tag{14}$$

where N represents the number of anchor nodes, $y^t_{i,j}$ represents the true label of whether a link exists between the node pair (v_i, v_j).

The overall loss consists primarily of the true label loss, combined with the two relational distillation losses described in Section 4.1. Therefore, the overall loss used for training the student MLP is as follows:

$$\mathcal{L} = \mathcal{L}_{sup} + \alpha \cdot \mathcal{L}_{rank} + \beta \cdot \mathcal{L}_{distribution}, \tag{15}$$

where, α, β control the relative importance of $\mathcal{L}_{rank}$ and $\mathcal{L}_{distribution}$, respectively.

5. Experiments

In this section, to demonstrate the effectiveness of KDCS-PPI, we have conducted extensive experiments to address the following key research questions:

- **RQ1:** How does KDCS-PPI perform in terms of both performance and efficiency compared to baseline methods?
- **RQ2:** How do different components affect the performance of KDCS-PPI?
- **RQ3:** How do hyperparameters impact model performance, and what are the potential reasons for their effects on the model?
- **RQ4:** Are the protein representations learned by the student MLP effective?

5.1. Experiment settings

Datasets. This study utilizes five datasets, namely STRING, SHS148k, SHS27k, A. thaliana, and S. cerevisiae, with STRING, SHS148k, and SHS27k arranged in decreasing order of scale. The STRING dataset comprises 14,952 proteins and 572,568 interaction pairs. Following the approach outlined in PIPR (Chen et al., 2019), proteins from the Homo sapiens subset of the STRING database with over 50 amino acids and less than 40% sequence similarity are selected to form two more arduous datasets, SHS148k and SHS27k, both of which are subsets of the STRING dataset. To increase the species diversity of the datasets, we also used two datasets from A. thaliana and S. cerevisiae (Yan et al., 2024). The former comes from the Arabidopsis thaliana interactome, representing plant interactions, while the latter comes from the Saccharomyces cerevisiae interactome, representing microbial interactions. Table 2 provides detailed information regarding the proteins and interaction relationships in each dataset. Besides, the 3D structures of each protein are predicted using AlphaFold2.

Evaluation metrics. We use two metrics, AUC and F1-score, to measure the performance of the model. The edges present in the PPI dataset are divided into training, validation, and test sets at a ratio of 6:2:2. The model is trained on the training set, and the model that performs best on the validation set is selected for evaluation on the test data. We have conducted five independent experiments using different random

Table 2
The statistics of the Five Datasets.

Dataset	# Proteins	# Interactions
STRING	14,952	572,568
SHS148k	5082	43,397
SHS27k	1663	7401
A. thaliana	1927	3339
S. cerevisiae	2666	6908

Table 3
Relevant Information on the Experimental Environment.

Experimental Environment	Relevant Information
Python	3.8
Pytorch	1.13.1
CPU	12x Xeon Platinum 8255C
GPU	RTX 3090
Operating System	Ubuntu 20.04

seeds, calculating the average values and standard deviations of the indicators, which are presented in the format of mean ±std.

Baselines. We categorize the comparison baselines into two classes. The first class is sequence-based methods, which include DNN-PPI (Li et al., 2018), PIRP (Chen et al., 2019), GNN-PPI (Lv et al., 2021), and SemiGNN-PPI (Zhao et al., 2023). The second class is structure-based methods, consisting of HIGH-PPI (Gao et al., 2023) and MAPE-PPI (Wu et al., 2024). Among them, HIGH-PPI is trained using only labeled data from downstream tasks, while MAPE-PPI employs unlabeled protein data for pre-training before further training with labeled data. Among these methods, GNN-PPI, SemiGNN-PPI, HIGH-PPI, and MAPE-PPI are designed to predict the type of interaction between proteins. We have made corresponding modifications to these models to adapt them to the PPI prediction task.

Implementation Details and Experimental Environment. We adopt a 4-layer GraphSAGE as the teacher model and a 3-layer MLP model as the student across all datasets. The margin hyperparameter δ in the ranking-based loss $\mathcal{L}_{rank}$ is set to 0.1. The temperature parameter τ, used to compute the predicted logit distributions $\mathbf{P}(C_i)$ and $\hat{\mathbf{P}}(C_i)$, is set to 1. The node-pair similarity threshold γ is defined as the median similarity among all node pairs in the datasets. We set the size of the candidate sample set k to 15, and the number of sampled positive/negative samples n to 9. The weights of the loss functions $\mathcal{L}_{rank}$ and $\mathcal{L}_{distribution}$, denoted as α and β, are both set to 1. Details of the hardware used for all experiments are provided in Table 3.

5.2. Performance comparison

In this section, to assess the performance and efficiency of the proposed method outlined in this paper against baseline methods, we have devised four experiments:

(1) A comprehensive performance comparison using AUC and F1-score on the STRING, SHS148k, and SHS27k datasets, followed by selecting the top-performing models for further comparison with the proposed method on the A. thaliana and S. cerevisiae datasets, to demonstrate the effectiveness of the proposed method across multiple species.
(2) Training models on small-scale datasets and evaluating them on large-scale ones to gauge generalizability.
(3) Assessing robustness by randomly removing edges from the PPI network at different ratios.
(4) On the SHS27k dataset, we compare the time efficiency of inference on the test set between the proposed model and the baseline methods.

5.2.1. Benchmark comparison

Experiments have been conducted for the method proposed in this paper and the baseline methods on the SHS27k, SHS148k, and STRING datasets. The results are presented in Table 4.

From the data in the table, we have the following observations:

(1) Among the sequence-based methods, GNN-PPI and SemiGNN-PPI achieve relatively better results. This may be because, compared to other sequence-based methods, these two methods use GNN to extract structural information from the PPI graph, which is beneficial for the PPI prediction task. Furthermore, when comparing the two methods, SemiGNN-PPI performs better, which may be attributed to its generation of dual views through operations on nodes and edges, and its combination with the Mean Teacher model on top of GNN. This allows the student model to better capture fine-grained structural information, thereby enhancing performance.

(2) HIGH-PPI, MAPE-PPI, and our model all leverage protein structural information to achieve comparatively superior performance. The results demonstrate that these approaches notably outshine others that solely depend on protein sequence details. This underscores the significance of protein structural data in enhancing model efficacy, as it offers more comprehensive biological insights than sequence data and is better suited for PPI prediction tasks.

(3) MAPE-PPI and our model achieve the best and second-best results, respectively, both outperforming other baseline methods. It is worth noting that both MAPE-PPI and our model employs unlabeled protein data for pre-training to obtain preliminary representations of proteins, which are then further optimized on the labeled PPI task. In contrast, other methods directly optimize the initial representations of proteins and the models on the labeled PPI task. This suggests that in PPI prediction tasks, the use of unlabeled protein data for pre-training can help the model learn more implicit features and interaction patterns, thereby enhancing model performance.

(4) KDCS-PPI achieves the best results across all three datasets. Compared to the second-best MAPE-PPI, KDCS-PPI incorporates relational knowledge distillation and counterfactual sampling strategy. This indicates that relation-level knowledge distillation can effectively utilize relational knowledge to enhance the model's predictive ability. Moreover, the counterfactual sampling strategy can also improve the model's performance to a certain extent, thereby offering more potential for PPI prediction tasks.

To further demonstrate the effectiveness of the proposed method across multiple species, we also conducted comparative experiments on two datasets from plants and microorganisms. In these experiments, we selected three top-performing baseline methods: MAPE-PPI, SemiGNN-PPI, and HIGH-PPI, and compared them with the proposed method on both datasets. The results are shown in Table 5.

From Table 5, we make the following observations:

(1) SemiGNN-PPI primarily relies on sequence information, and its relatively poor performance on both datasets indicates that solely relying on sequence information is insufficient to capture complex protein representations. In contrast, HIGH-PPI and MAPE-PPI incorporate structural information, leading to a significant performance improvement. This suggests that methods integrating structural information are more effective for protein representation learning, thereby enhancing the accuracy of PPI prediction.

(2) Compared to their performance on the S. cerevisiae dataset, all models exhibit varying degrees of performance decline on the A. thaliana dataset. This may be attributed to the smaller dataset size of A. thaliana compared to S. cerevisiae, indicating that factors such as dataset size and network structure can influence prediction performance.

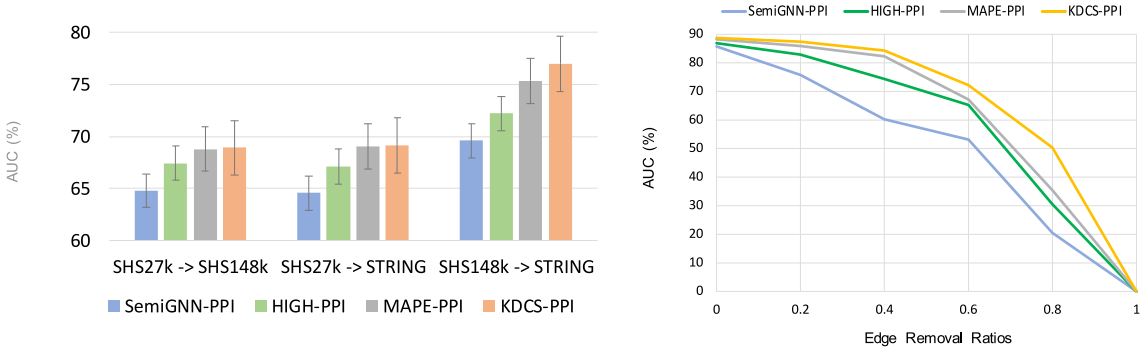
(3) Our method remains superior on both datasets, demonstrating its effectiveness in predicting PPIs across different species. Combined with its performance on the STRING dataset and its subsets, this indicates that our model exhibits strong generalization capability across datasets of different scales and across species.

Table 4
Comparison of the Overall Performance on STRING and Its Subsets.

Method	SHS27k		SHS148k		STRING	
	AUC (%)	F1-score (%)	AUC (%)	F1-score (%)	AUC (%)	F1-score (%)
DPPI	74.29 ± 1.24	66.87 ± 0.95	82.93 ± 0.48	73.23 ± 0.61	90.52 ± 0.59	82.09 ± 1.30
DNN-PPI	77.34 ± 0.61	69.69 ± 0.49	84.34 ± 0.82	73.17 ± 0.72	89.77 ± 0.73	82.91 ± 0.41
PIPR	80.07 ± 0.43	71.49 ± 0.72	87.85 ± 1.28	79.21 ± 1.04	92.96 ± 0.25	84.51 ± 0.79
GNN-PPI	83.81 ± 1.20	74.30 ± 0.52	89.23 ± 0.47	81.41 ± 0.93	94.20 ± 0.62	85.21 ± 0.85
SemiGNN-PPI	85.75 ± 0.74	78.42 ± 0.34	90.88 ± 1.25	82.69 ± 1.20	94.08 ± 0.48	86.82 ± 0.62
HIGH-PPI	86.91 ± 0.98	80.23 ± 0.82	91.32 ± 0.64	85.24 ± 1.32	–	–
MAPE-PPI	88.23 ± 0.57	81.18 ± 0.31	93.14 ± 0.60	88.02 ± 0.94	97.72 ± 0.75	90.42 ± 0.81
KDCS-PPI	**88.72 ± 0.78**	**81.89 ± 0.23**	**94.42 ± 0.56**	**89.38 ± 0.67**	**98.12 ± 0.34**	**91.19 ± 0.76**

Table 5
Comparison of the Overall Performance on A. thaliana and S. cerevisiae.

Method	A. thaliana		S. cerevisiae	
	AUC (%)	F1-score (%)	AUC (%)	F1-score (%)
SemiGNN-PPI	78.23 ± 0.39	71.23 ± 0.40	78.72 ± 0.42	74.21 ± 0.68
HIGH-PPI	79.26 ± 0.46	74.32 ± 0.33	81.59 ± 0.64	74.85 ± 0.51
MAPE-PPI	82.93 ± 0.52	77.23 ± 0.74	84.74 ± 0.57	78.21 ± 0.42
KDCS-PPI	**83.01 ± 0.39**	**77.92 ± 0.43**	**85.12 ± 0.67**	**79.72 ± 0.36**



(a) Generalizability (b) Robustness

Fig. 4. Comparison of KDCS-PPI with other methods on the SHS27k dataset. (a) Assessing generalizability by applying models trained on small-scale data to large-scale datasets. The x-axis represents different generalization scenarios, and the y-axis indicates AUC values. (b) Evaluating robustness by randomly deleting a certain proportion of edges in the PPI graph on the SHS27k dataset. The x-axis represents different edge removal ratios, and the y-axis indicates AUC value.

5.2.2. Generalization analysis.

To evaluate the generalizability of KDCS-PPI, we train the model using a small-scale dataset and then apply it to a large-scale dataset to observe changes in its performance. Specifically, we train the model using the SHS27k dataset and then apply it to the SHS148k and STRING datasets, as well as train the model using the SHS148k dataset and then apply it to the STRING dataset. We only select three models that perform better as indicated in Section 5.2.1 — MAPE-PPI, HIGH-PPI, and SemiGNN-PPI — as comparison baselines. The results are presented in Fig. 4(a).

From Fig. 4(a), we have the following observations:

(1) After training the model on the SHS27k dataset, for SemiGNN-PPI and HIGH-PPI, their generalizability on the SHS148k dataset is superior to that on the STRING dataset. However, for MAPE-PPI and KDCS-PPI, the generalizability performance in these two scenarios is more stable, indicating that using unlabeled protein information for pre-training can capture more comprehensive protein information, thereby enhancing the model's performance across various generalization scenarios.

(2) Further analysis reveals that compared to training on SHS27k, models trained on the SHS148k dataset show a greater performance improvement. This indicates that a larger dataset size is more beneficial for enhancing the model's generalization ability. Particularly noteworthy is that KDCS-PPI exhibits a more significant performance improvement when the amount of data increases. This phenomenon suggests that KDCS-PPI can more effectively learn deep-level features and patterns from large-scale data. This might be due to the fact that, on the SHS148k dataset, the counterfactual sampling strategy has a broader search space to select more reliable positive/negative samples, thereby improving the model's performance.

(3) Across all generalization scenarios, KDCS-PPI has demonstrated excellent adaptability, achieving performance comparable to or better than the baseline methods. This indicates that KDCS-PPI holds a competitive advantage in real-world applications and is better equipped to handle the challenges of generalization scenarios in practice.

5.2.3. Robustness analysis.

To evaluate the robustness of our model, we perform random edge deletion on the PPI network in the SHS27k dataset at ratios of 0, 0.2, 0.4, 0.6, 0.8, and 1, respectively, to simulate scenarios where protein-protein interactions are not fully explored in the PPI network. Similar to Section 5.2.2, we only select MAPE-PPI, HIGH-PPI, and SemiGNN-PPI as the comparison baselines. The results are presented in Fig. 4(b).

Through the experimental data, several observations can be made:

(1) Under different edge deletion ratios, the performance of KDCS-PPI consistently surpasses that of the baseline methods, indicating that KDCS-PPI has superior robustness compared to the baseline methods. Particularly noteworthy is that when the edge deletion ratio is less than 0.4, the decline in model performance is very slight, suggesting that KDCS-PPI can to some extent overlook the impact caused by insufficient

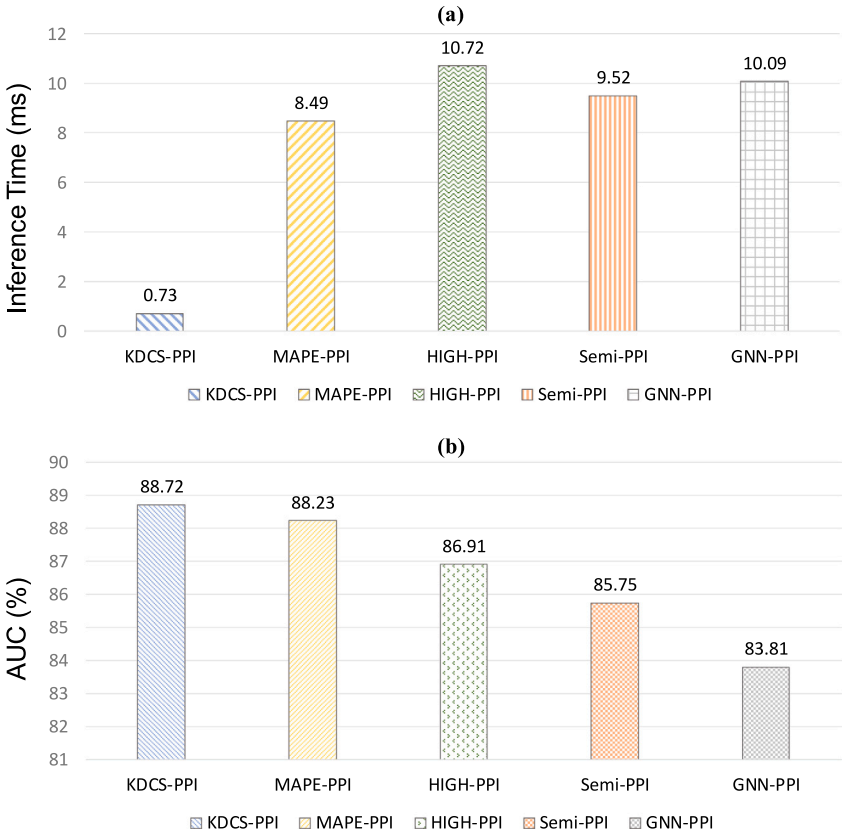


Fig. 5. Comparison of KDCS-PPI with some baseline methods in terms of time efficiency and effectiveness on the SHS27k dataset.

exploration of protein-protein interactions in the PPI network, thereby better adapting to real-world application scenarios.

(2) Additionally, in scenarios with a lower edge deletion ratio, the advantage of KDCS-PPI over other models is relatively small. However, as the edge deletion ratio increases, the advantage of the proposed model becomes more pronounced, particularly when the edge deletion ratio reaches 0.8, where KDCS-PPI achieves the greatest performance advantage compared to other baseline methods. We speculate that the possible reason for this phenomenon is that, unlike other models that use GNN for prediction, our model employs an MLP that does not rely on data structure, which to some extent enhances the model's ability to adapt to edge deletion situations.

5.2.4. Time efficiency analysis.

To evaluate the time efficiency and effectiveness of KDCS-PPI, we have compared it with several baseline methods in terms of time efficiency and effectiveness during the inference process. First, the models are trained on the SHS27k training set, and then the trained models are used to predict protein-protein interactions on the test set. The time required for prediction and the AUC metric are recorded. The time comparison is displayed in Fig. 5(a), and the AUC metric comparison is shown in Fig. 5(b).

From the data presented in the figures, the following observations can be made:

(1) KDCS-PPI demonstrates significantly higher time efficiency while maintaining a superior AUC performance compared to other methods. Specifically, the improvement over the HIGH-PPI method is the most pronounced, with a 14.68-fold increase in time efficiency, showcasing a clear advantage. Additionally, when compared to MAPE-PPI, which has a performance similar to the proposed model, the time efficiency of the proposed model is 11.63 times faster. The reason for this improvement in time efficiency is that other methods rely on GNN for inference, which necessitates the handling of complex graph structures, resulting in substantial time overhead. In contrast, the proposed model employs an MLP for inference, which does not depend on graph structures, thus allowing for a more concise and efficient inference process.

(2) Concurrently, we can observe that the proposed model maintains superior performance compared to other baseline methods while enhancing time efficiency. This suggests that through relation-level knowledge distillation, the proposed model effectively transfers graph structure information from the teacher GNN model to the student MLP model, allowing it to capture important feature information without relying on graph structures. This design enables the proposed model to have a relative advantage in both time efficiency and performance, making it particularly suitable for practical application scenarios involving large-scale PPI data.

5.3. Hyperparametric sensitivity.

In this section, the primary focus is on studying the impact of three key hyperparameters in the counterfactual sampling strategy: the number of candidate samples, denoted as k, and the number of samples to be sampled, denoted as n, and the hyperparameters related to the teacher networks. To facilitate the discussion, we replace the direct use of n with the sampling ratio n/k for the analysis of n. Furthermore, to delve deeper into the underlying reasons for the impact of these two hyperparameters on the model, we conduct an in-depth analysis of the distribution of the number of samples sampled through counterfactual sampling within the candidate samples. Additionally, to investigate the impact of the model architecture and the number of layers in the teacher networks on performance, we design corresponding analytical experiments.

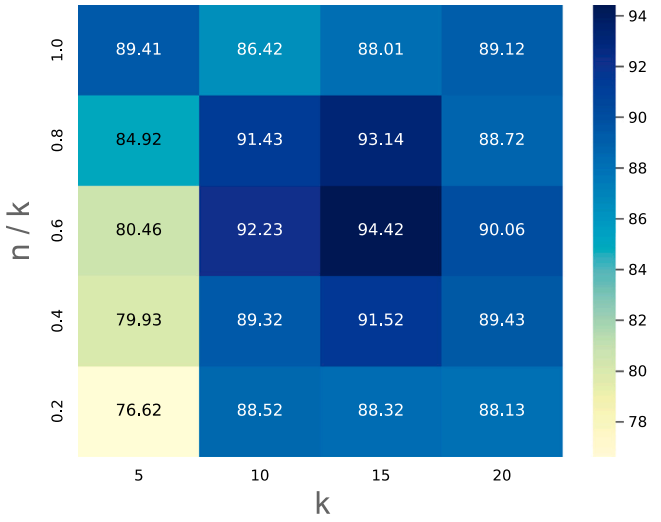


Fig. 6. Sensitivity analysis of the hyperparameters k and n/k on the SHS148k dataset. The x-axis represents the number of candidate positive samples, while the y-axis denotes the ratio of sampled samples to candidate samples, n/k. The numbers in each cell of the heatmap indicate the model's AUC score under the current hyperparameters n and n/k.

5.3.1. Size of the candidate sample set and number of selected samples.

To investigate the effect of hyperparameters on the model, we have examined the sensitivity of the model to the number of candidate positive/negative samples k and the ratio n/k sampled positive/negative samples to the number of candidates on the SHS148k dataset. During the experiments, we have found that for negative samples, sufficient samples could always be identified using counterfactual node selection. This is because the number of unlinked node pairs in the PPI graph is significantly greater than the number of linked node pairs. Therefore, only the sampling of positive samples has been analyzed. On the SHS148k dataset, we have adopted four candidate sample sizes k={5, 10, 15, 20} and four sampling ratios n/k={0.2, 0.4, 0.6, 0.8, 1}, and their corresponding AUC values are recorded. The results are shown in Fig. 6.

From the experimental results, it can be seen that:

(1) When the number of candidate samples is 10 or 15 and the sampling ratio is between 0.4 and 0.8, the model's performance remains relatively stable, indicating that our model is, to some extent, insensitive to hyperparameters. The reason why this hyperparameter setting can achieve better performance may be that the model is able to sample positive/negative samples from the constructed candidates that genuinely contributed to model training. This also demonstrates the effectiveness of the counterfactual sampling strategy.

(2) When the number of candidate samples and the number of sampled samples are either too small or too large, the model's performance declines. The potential reason for this is that if the number of sampled samples is too small, the training data would be insufficient, making it difficult for the model to effectively capture the complex patterns and features within the relation-level knowledge, leading to poor training performance. On the other hand, when the number of candidate samples and the number of sampled samples are too large, due to the candidate sample construction strategy, some nodes would be repeatedly selected into the candidate set. This would cause the model to frequently sample duplicate data during training, making it difficult for the model to learn truly useful information, thus limiting its performance. Therefore, it is crucial for our model to select an appropriate number of candidate samples k and sampled samples n.

5.3.2. Distribution of counterfactual samples.

To further analyze the influence of the number of candidate samples, i.e., k, for each anchor node that is selected based on the counterfactual sampling strategy, we have constructed candidate sample sets for each node in the SHS27k, SHS148k, and STRING datasets. Each anchor node v_i has a candidate sample set $\mathbf{M}_i$, which includes 15 positive samples and 15 negative samples. We then calculate the number of counterfactual samples within each $\mathbf{M}_i$ and analyze the distribution of anchor nodes based on the number of counterfactual samples in their candidate sets. It is important to note that, due to the sparse nature of PPI graph, all negative samples in the candidate sets could be obtained through the counterfactual sampling strategy. Thus, the analysis focuses solely on the distribution of anchor nodes corresponding to the number of counterfactual positive samples. The results are shown in Fig. 7.

From the above data, it can be observed that:

(1) Although the proportion of anchor nodes with zero counterfactual positive samples in the candidate sample sets is the highest across all three datasets, this proportion gradually decrease with the increasing size of the dataset—from 41% in the SHS27k dataset to 20% in the STRING dataset. This indicates that, in smaller datasets, many anchor nodes lack sufficient contextual information to generate counterfactual positive samples, but this issue tends to be alleviated as the dataset size increases.

(2) Moreover, it can be observed that as the dataset size increases, an increasing number of anchor nodes are able to obtain more counterfactual positive samples. This trend indicates that, in larger datasets, the counterfactual sampling strategy becomes more effective at selecting suitable positive samples, thereby providing more information for model training.

(3) On all three datasets, more than half of the nodes are able to find at least one counterfactual positive sample through the counterfactual sampling strategy. This indicates that regardless of the dataset size, the counterfactual sampling strategy consistently provides a substantial amount of reliable data for model training, which is beneficial for the model's training process.

5.3.3. Ablation study.

To further analyze the effects of the two key components in our proposed methods — the relational knowledge distillation strategy and the counterfactual sampling — we have designed five variants of KDCS-PPI for comparison with the full model:

- $w/o\ \mathcal{L}_{rank}$: Removes the ranking-based matching strategy from KDCS-PPI.
- $w/o\ \mathcal{L}_{distribution}$: Removes the distribution-based matching strategy from KDCS-PPI.
- $w/o\ CF$: Replaces the counterfactual sampling strategy with random sampling during model training.
- $w/o\ \mathcal{L}_{rank}, CF$: Removes the ranking-based matching strategy from KDCS-PPI and replaces counterfactual sampling with random sampling.
- $w/o\ \mathcal{L}_{distribution}, CF$: Removes the distribution-based matching strategy from KDCS-PPI and replaces counterfactual sampling with random sampling.

Experiments have been conducted on the SHS27k, SHS148k, and STRING datasets, and the results are presented in Table 6.

It can be observed that:

(1) When each component of the model removes, the model's performance declines to varying degrees, indicating that each component is crucial for the training and optimization of the model. In particular, the performance drops the most when the distribution-based matching strategy is removed. This suggests that fine-grained logit scores are more important than ranking information in the model optimization

Table 6
Ablation study of different components of KDCS-PPI on three datasets.

Method	SHS27k	SHS148k	STRING	Average
$w/o\ \mathcal{L}_{rank}$	86.92	92.28	92.52	90.57(−2.75)
$w/o\ \mathcal{L}_{distribution}$	86.01	91.47	92.01	89.83(−3.49)
$w/o\ CF$	88.40	93.95	95.87	92.74(−0.58)
$w/o\ \mathcal{L}_{rank}, CF$	86.11	91.73	92.13	89.99(−3.33)
$w/o\ \mathcal{L}_{distribution}, CF$	85.43	90.86	91.57	89.28(−4.04)
KDCS-PPI	88.72	94.42	96.82	93.32

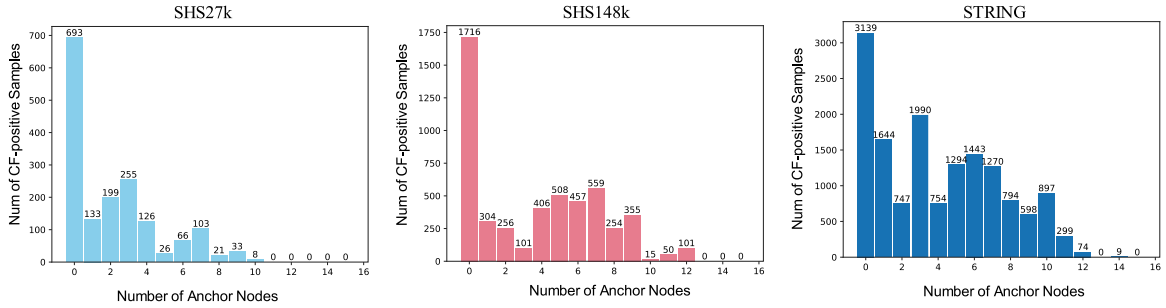


Fig. 7. Distribution of the number of counterfactual samples in the candidate set across the three datasets. The x-axis represents the number of counterfactual positive samples in the candidate sample set, while the y-axis indicates the corresponding number of anchor nodes.

process. This may be because logit scores provide more accurate prediction values, allowing the model to adjust parameters and learn features more effectively, thereby improving overall performance.

(2) In addition, the importance of the counterfactual sampling strategy becomes increasingly prominent with the increase in the number of samples. This might be because, with a smaller number of samples, it is challenging to locate enough samples that meet the criteria for counterfactual sampling, leading to suboptimal utilization of the data for training. As the number of samples increases, the number of samples that meet the counterfactual sampling criteria also grows, providing the model with more reliable samples and thereby enhancing its performance. This indicates that nodes selected through the counterfactual sampling strategy can more effectively contribute to model optimization, and as the dataset size grows, the counterfactual sampling strategy becomes increasingly supportive of performance improvement.

5.3.4. Hyperparameter analysis of teacher network design.

To investigate the impact of the GNN architecture and the number of layers used in the teacher network on model performance, we conducted experiments on the SHS27k dataset using different GNN architectures and layer numbers. The results are presented in Fig. 8.

Through observation, we can make the following findings:

(1) When the number of layers is set to 4, all architectures perform relatively well, with the GraphSAGE model achieving the best performance. This suggests that the GraphSAGE architecture is particularly effective when the network depth is moderate, and it is able to extract meaningful features at this layer level, outperforming other models.

(2) The performance of the GraphSAGE model remains relatively stable across the layer configurations of 3, 4, and 5. This indicates that the GraphSAGE model is less sensitive to variations in the number of layers within this range, suggesting its robustness in learning and generalizing across different network depths.

Based on the above analysis, the GraphSAGE model demonstrates relative stability and achieves optimal performance with 4 layers. Therefore, we select the 4-layer GraphSAGE model as the teacher network.

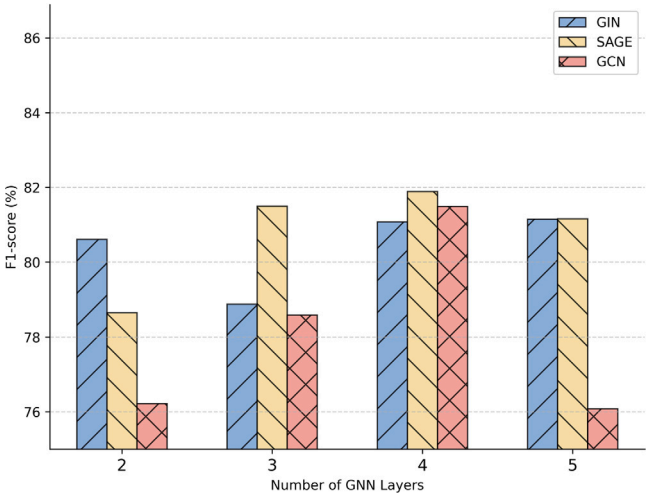


Fig. 8. Hyperparameter analysis of teacher network design. Different colors represent various GNN architectures, with the x-axis indicating the number of layers used in each architecture and the y-axis representing the model's F1 score.

5.4. Latent space visualization.

To visually demonstrate the quality of protein representations learned by the student MLP through knowledge distillation, we use T-SNE (Van der Maaten & Hinton, 2008) to reduce the dimensionality of the protein representations learned by the student MLP to two dimensions for visualization on the SHS27k dataset. We selected 23 protein families from the SHS27k dataset, each containing more than seven proteins. Proteins within these families are displayed using different colors, while all other proteins are shown in gray. The visualization results are presented in Fig. 9.

From the figure, it can be observed that the representations learned by the student MLP effectively cluster proteins from the same family together to some extent, while separating proteins from different families. In particular, the model excels at clustering proteins from the Small GTPase superfamily, Intermediate filament family,



Fig. 9. Visualization of protein representations learned by the student MLP on the SHS27k dataset.

G-protein coupled receptor 1 family, and Type I cytokine receptor family, while distinguishing between different protein families. This ability to cluster representations of proteins from the same family and separate those from different families further validates the effectiveness of the proposed model and its potential application in the field of bioinformatics.

6. Conclusion and future work

In this paper, we have proposed a knowledge distillation method with counterfactual node selection for the PPI prediction task. Two relational distillation strategies are designed to transfer the knowledge learned by the teacher GNNs through the PPI network to the student MLP, thereby combining the effectiveness of GNNs with the efficiency of MLP. Furthermore, during the knowledge distillation process, a counterfactual sampling strategy is employed to select more reliable samples for anchor nodes, optimizing model training. Compared to state-of-the-art methods, our approach demonstrates strong performance in both effectiveness and efficiency. However, we also note that the counterfactual strategy does not always provide sufficient positive/negative samples for all anchor nodes. Optimizing this strategy remains a valuable direction for future research.

CRediT authorship contribution statement

Bin Deng: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Huifang Ma:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Ruijia Zhang:** Validation, Investigation, Writing – review & editing. **Zhixin Li:** Resources, Writing – review & editing. **Liang Chang:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my data/code at the https://github.com/bin-db/KDCS-PPI.

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